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ASCEND – Advanced Support for Career Exploration and Navigating Directions(SIH1666)

Abstract

Career counselling and guidance are crucial for students at every stage of their academic journey, helping them make informed decisions about their future. However, many students and families in India lack access to proper career guidance, leading to choices based on limited information or societal pressures. This can result in mismatched skills, job dissatisfaction, and even unemployment. To address this, the National Education Policy (NEP) 2020 emphasizes the importance of guiding students toward suitable career paths by aligning their education with their career goals.

This paper proposes an innovative solution that includes AI-powered platforms, mentorship programs, interactive tools, and resource portals designed to assist students from middle school to postgraduate levels in exploring and choosing careers that match their interests and strengths. These tools will empower students to explore various career opportunities, develop necessary skills, and plan their education accordingly.

Key components of the solution include:

A) **AI-Powered Career Guidance Platforms**:
These platforms will provide personalized career suggestions based on students' unique interests, strengths, and emerging job trends.

B) **Career Mentorship Programs**:
Students will connect with professionals from their chosen fields, giving them real-world insights and guidance.

C) **Interactive Career Exploration Tools**:
Engaging, digital tools will allow students to explore different careers through virtual simulations, videos, and interactive activities.

D) **Career Resource Portals**:
An extensive online portal will offer resources such as career paths, required skills, educational plans, and job market trends.

This solution supports NEP 2020's goal of helping students make informed career choices through personalized guidance, technology-driven tools, and mentorship programs, ensuring that their education aligns with their career aspirations at every academic level.

Modules identified:

1. AI-Powered Career Guidance Platform
2. Career Mentorship Program
3. Interactive Career Exploration Tools
4. Career Resource Portal

Features in each Module :

Modules	Features	Description	AI/ML Models & Technologies
AI-Powered Career Guidance Platform	1-Personalized Career Recommendations	Adapts recommendations based on evolving job market trends and skill demands.	Collaborative Filtering Algorithms, Machine Learning Models
	2-Interest and Aptitude Assessment	Provides online assessments (psychometric tests, aptitude tests). AI interprets results to match students with suitable career paths.	Psychometric Analysis, Machine Learning Models
	3-Real-Time Job Market Insights	Uses NLP to analyze job market data and provide real-time insights into trending skills, professions, and job opportunities.	NLP Models (SpaCy, BERT, GPT), Data Mining
	4-Career Pathway Visualizations	Generates visual roadmaps outlining the qualifications, skill sets, and steps required to pursue chosen career paths.	Knowledge Graphs, Visualization Tools
Career Mentorship Program	5-AI-Based Mentor Matching	Matches students with mentors based on interests, skills, and experience using AI recommendation algorithms.	Recommendation Algorithms, Collaborative Filtering
	6-Mentor-Student Interaction Portal	Facilitates video calls, chats, and feedback between students and mentors.	Chatbot AI, Video Conferencing Tools
	7-Progress Tracking and Feedback	ML algorithms track students' progress, monitor skill acquisition, and provide personalized feedback from mentors.	Machine Learning Models, Progress Monitoring Systems
	8-Learning Resources and Guidance	AI curates relevant learning materials and resources based on mentor discussions and career goals.	NLP Summarization, Resource Recommendation Systems

Interactive Career Exploration Tools	9-Startup Ideas	Designed to inspire and guide users who are interested in exploring entrepreneurial opportunities aligned with their career aspirations.	API (Wikipedia and NewsAPI)
	10- Public Pathways	The goal is to guide users toward opportunities and roles available in various government sectors	Front-End Technologies (HTML,CSS,JS)
	11-Chatbot	It is an AI-powered virtual assistant designed to guide users through the Career Dendrogram platform.	API (Gemini), bootstrap
	12-Dynamic Career Path Exploration	ML models analyze student preferences and suggest dynamic pathways based on real-world transitions.	Machine Learning, Career Path Recommendation Systems
Career Resource Portal	13-Curated Career Resources	AI recommends articles, guides, and videos based on a student's profile and interests.	NLP-based Summarization, Recommendation Engines
	14-Search and Filter Options	NLP-powered search engine allows students to filter career information, educational pathways, required skills, and certifications.	NLP (SpaCy, GPT), Search Algorithms
	15-Market Trend Analysis	Analyzes job markets and provides detailed reports on career demand, skill gaps, and opportunities.	AI Analytics, Data Mining, Market Trend Models
	16-Educational Pathways Mapping	AI-driven roadmap outlining education, certifications, internships, and skills required for specific career goals.	Knowledge Graphs, Pathway Recommendation Engines
	17-Skill Assessment and Learning Plans	ML models analyze current skills and recommend personalized learning plans for skill development.	ML Algorithms (Scikit-learn, XGBoost), Skill Matching Algorithms

Signature of Guide

Signature of Project Co-ordinator